

MANAGUA INTL, Nicaragua

WMO: 787410

Lat: 12.141N

Lon: 86.168W

Elev: 59

StdP: 100.62

Time Zone: -6.00 (W06)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	20.0	20.9	14.8	10.6	28.0	15.8	11.3	28.5	9.8	31.2	8.9	30.9	0.6	110	0.359

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.1	36.0	24.4	35.2	24.2	34.8	24.2	26.6	31.5	26.3	31.2	26.1	31.0	4.8	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.5	28.4	25.1	20.3	28.3	24.9	20.1	28.2	83.2	31.7	82.0	31.6	81.0	31.4	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.9	7.0	DB	17.7	37.0	1.8	0.5	16.4	37.4	15.3	37.7	14.3	38.0	12.9	38.4	(4)
(5)			WB	15.4	27.6	1.4	0.8	14.4	28.2	13.6	28.7	12.9	29.1	11.8	29.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	26.9	27.5	28.5	29.6	29.4	28.2	28.0	28.2	27.9	27.4	27.0	26.9	(6)
(7)		DBStd	1.36	0.99	1.02	1.17	1.04	1.37	1.04	0.96	0.91	0.95	1.03	0.96	1.04	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6557	524	491	573	589	600	545	558	565	537	539	510	524	(10)
(11)		CDD18.3	3515	266	258	315	339	342	295	299	307	287	281	260	266	(11)
(12)		CDH23.3	37415	2681	2794	3824	4466	4153	3017	2976	3107	2758	2573	2466	2601	(12)
(13)		CDH26.7	16495	1146	1280	1917	2373	2031	1185	1166	1299	1095	943	968	1091	(13)
(14)	Wind	WSAvg	2.9	3.7	4.0	4.2	3.9	3.0	2.3	2.7	2.4	1.9	1.8	2.1	3.0	(14)
(15)	Precipitation	PrecAvg	1107	5	0	1	6	113	207	133	155	206	213	57	11	(15)
(16)		PrecMax	1743	17	2	7	32	226	422	267	362	367	430	203	37	(16)
(17)		PrecMin	671	0	0	0	0	3	82	50	40	70	64	4	0	(17)
(18)		PrecStd	292	5	1	2	10	74	84	61	76	82	107	51	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	35.0	36.1	37.0	36.9	34.9	34.2	35.0	34.2	33.2	33.2	33.9	(19)
(20)			MCWB	23.4	24.0	24.3	24.5	24.7	25.3	25.0	25.5	25.6	25.7	24.8	24.3	(20)
(21)		2%	DB	33.1	34.0	35.1	36.1	35.9	33.8	33.2	34.0	33.3	32.9	32.2	33.0	(21)
(22)			MCWB	23.1	23.4	23.8	24.3	24.5	25.5	25.0	25.4	25.6	25.6	24.5	23.8	(22)
(23)		5%	DB	32.2	33.1	34.2	35.2	35.0	32.9	32.8	33.0	32.9	32.0	32.0	32.1	(23)
(24)			MCWB	22.9	23.1	23.3	24.0	24.5	25.4	25.1	25.4	25.6	25.3	24.4	23.6	(24)
(25)		10%	DB	31.8	32.5	33.9	34.9	34.0	32.0	31.9	32.1	31.9	31.1	31.1	31.5	(25)
(26)			MCWB	22.8	22.9	23.2	23.9	24.6	25.3	25.0	25.2	25.4	25.4	24.4	23.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.8	25.6	25.8	26.5	26.7	26.7	26.6	26.7	26.7	26.2	25.6	(27)	
(28)			MADB	31.2	32.1	33.3	32.4	32.1	31.4	31.0	31.8	31.3	30.9	30.8	30.7	(28)
(29)		2%	WB	24.2	24.6	25.0	25.8	26.2	26.4	26.1	26.2	26.4	26.2	25.8	24.9	(29)
(30)			MADB	30.7	31.4	32.4	32.3	31.5	31.1	30.8	31.4	31.1	30.6	30.4	30.5	(30)
(31)		5%	WB	23.7	24.0	24.5	25.3	25.9	26.1	25.7	26.0	26.1	26.0	25.4	24.4	(31)
(32)			MADB	30.2	30.9	31.8	31.8	31.1	30.7	30.4	31.2	30.9	30.3	30.0	30.0	(32)
(33)		10%	WB	23.3	23.5	23.9	24.9	25.5	25.8	25.4	25.6	25.8	25.7	25.0	24.0	(33)
(34)			MADB	29.6	30.4	31.2	31.4	30.7	30.2	30.0	30.6	30.4	29.8	29.6	29.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.5	11.0	11.4	11.1	9.5	8.0	8.3	8.9	8.6	7.9	9.1	10.1	(35)
(36)			MCDBR	11.0	11.5	11.7	11.4	10.8	9.1	9.3	9.7	9.7	9.0	9.9	10.5	(36)
(37)			MCWBR	3.6	3.7	3.5	3.2	3.0	3.2	3.2	3.4	3.4	3.5	3.8	3.6	(37)
(38)		5% WB	MCDBR	10.4	11.0	11.2	10.7	9.2	8.1	8.3	9.0	8.7	8.2	8.7	9.8	(38)
(39)			MCWBR	3.8	4.0	3.9	3.4	3.1	3.3	3.3	3.5	3.5	3.5	3.7	3.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.366	0.371	0.380	0.453	0.499	0.446	0.448	0.443	0.427	0.405	0.377	0.371	(40)	
(41)		taud	2.555	2.538	2.501	2.277	2.173	2.361	2.351	2.360	2.403	2.489	2.572	2.569	(41)	
(42)		Ebn at Noon	938	944	937	861	808	842	843	858	880	898	920	924	(42)	
(43)		Edh at Noon	100	106	112	139	151	124	125	126	121	110	98	97	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.30	5.90	6.48	6.29	5.39	5.13	5.18	5.46	5.32	4.95	5.08	5.03	(44)	
(45)		RadStd	0.18	0.24	0.21	0.28	0.31	0.25	0.28	0.23	0.38	0.45	0.23	0.22	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	+0.34	+0.57	N/A	+0.16	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page